

**Title:**

✓ Build a Production-Grade Algorithmic Trading System for SPX 0DTE Spreads (IBKR API, Excel Dashboard, Backtesting)

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**Description:**

We are looking for an experienced algorithmic trading system developer to help design, build, and deploy a robust, production-ready options trading system focused on SPX 0DTE spread strategies.

This is a comprehensive, multi-component project involving strategy backtesting, real-time execution via the Interactive Brokers (IBKR) API, advanced risk controls, event-driven trade logic, integration with a delta-hedging module, and a full Excel-based monitoring dashboard.

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 **Scope of Work:****1. Strategy Design & Backtesting**

- Backtest multiple SPX 0DTE spread and TLT wheel strategies on QuantConnect.
- Optimize parameters (e.g., strike distance, entry time, stop loss) to maximize Sharpe ratio and minimize drawdown.

**2. Live Trading via IBKR API**

- Place short vertical spreads based on real-time Greeks, volatility, and probabilities.
- Slippage-aware execution with fallback mechanisms.
- Dynamic position sizing with strict \$15,000 daily risk cap.

**3. Event-Driven Trade Module**

- Detect and fade sudden directional SPX moves.
- Logic for quick reversion-based spreads.

**4. Delta Hedging Integration**

- Refactor existing delta hedger.
- Maintain neutral portfolio delta using futures or options.

#### 5. **Excel-Based Dashboard**

- Live monitoring of trades, capital usage, risk metrics, Greeks, and performance.
- Control panel for strategy inputs and diagnostics.

#### 6. **Monitoring & Risk Management**

- Real-time P&L tracking, automated exits on defined thresholds.
- Live alerts, safety checks, and strict risk compliance.

#### 7. **Error Handling & Stability**

- Handle order rejections, API disconnects, and fill anomalies.
- Implement retries, logging, and safe shutdown mechanisms.

#### 8. **Testing, Deployment, and Post-Launch Support**

- Rigorous testing in IBKR paper account.
- Smooth transition to live trading once validated.
- Ongoing support for monitoring, tuning, and enhancements.



#### **Timeline:**

Up to **3 months**, including testing, optimization, integration, and rollout.

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#### **Ideal Candidate:**

- Proven experience building algorithmic trading bots (ideally for options)
- Strong with IBKR API, QuantConnect, and Excel/VBA integration

- Skilled in risk management, robust execution logic, and fault-tolerant design
  - Communicative, detail-oriented, and willing to collaborate on iteration
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 **To Apply:**

Please share:

- Relevant past projects (especially in options or IBKR)
- Your approach to backtesting and live trade risk management
- Estimated time and cost
- Questions or suggestions to refine the system further

We're looking for someone long-term who can evolve this into a resilient, intelligent trading engine. If you're passionate about financial automation, we want to hear from you!